positive (Figure 1.7). Here, the 5th division of the head scale coincides with the pitch scale axis. Then the zero error is positive and is given by,

Z.E = + (n × LC) where 'n' is the head scale coincidence. In this case, Zero error = + (5 × 0.01) = 0.05mm. So the zero correction is – 0.05 mm.

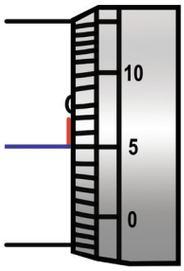



**Figure 1.7** Positive Zero Error

### Negative zero error

When the plane surface of the screw and the opposite plane stud on the frame are brought into contact, if the zero of the head scale lies above the pitch scale axis, the zero error is negative (Figure 1.8). Here, the 95$^{th}$ division coincides with the pitch scale axis. Then the zero error is negative and is given by,

$$ZE = -(100 - n) \times LC$$
$$ZE = -(100 - 95) \times LC$$
$$= -5 \times 0.01$$
$$= -0.05 \text{ mm}$$

The zero correction is + 0.05mm.

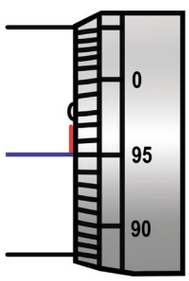



**Figure 1.8** Negative Zero Error

**Activity 2**

Determine the thickness of a single sheet of your science textbook with the help of a Screw gauge.

## 1.8 Measuring Mass

We commonly use the term 'weight' which is actually the 'mass'. Many things are measured in terms of 'mass' in the commercial world. The SI unit of mass is kilogram (kg). In any case, the units are based on the items purchased. For example, we buy gold in gram or milligram, medicines in milligram, provisions in gram and kilogram and express cargo in tonnes.

Can we use the same instrument for measuring the above listed items? Different measuring devices have to be used for items of smaller and larger masses. In this section we will study about some of the instruments used for measuring mass.




The shell of an egg is 12% of its mass. A blue whale can weigh as much as 30 elephants and it is as long as 3 large tour buses.

### Common (beam) balance

A beam balance compares the sample mass with a standard reference mass (Standard reference masses are 5 g, 10 g, 20 g, 50 g, 100 g, 200 g, 500 g, 1 kg, 2 kg, 5 kg). This balance can measure mass accurately up to 5 g (Figure 1.9).



**Figure 1.9** Common beam balance

### Physical balance

This balance is used in labs and is similar to the beam balance but it is a lot more sensitive and can measure mass of an object correct to a milligram (Figure 1.10).



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